

HA NOI, Viet Nam

WMO: 488200

Lat: 21.0205N

Lon: 105.8070E

Elev: 39

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 49.9	(c) 51.9	(d) 35.7	(e) 30.7	(f) 58.3	(g) 39.3	(h) 35.4	(i) 58.3	(j) 16.9	(k) 64.3	(l) 15.7	(m) 64.6	(n) 6.0	(o) 50	(p) 0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 13.6	(c) 97.2	(d) 81.3	(e) 95.3	(f) 81.2	(g) 93.5	(h) 81.2	(i) 84.8	(j) 90.7	(k) 84.0	(l) 89.9	(m) 83.1	(n) 89.1	(o) 6.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
84.0	178.7	87.4	82.6	170.4	86.7	82.1	167.2	86.4	49.4	90.2	48.4	89.5	47.3	89.2	95.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.7	(b) 13.7	(c) 12.1	(e) 46.4	(f) 101.6	(g) 2.6	(h) 2.4	(i) 44.5	(j) 103.3	(k) 43.0	(l) 104.7	(m) 41.5	(n) 106.0	(o) 39.6	(p) 107.8		
(a) 43.2	(b) 87.0	(c) 2.4	(e) 43.2	(f) 87.0	(g) 2.4	(h) 2.4	(i) 41.5	(j) 88.7	(k) 40.1	(l) 90.1	(m) 38.7	(n) 91.4	(o) 37.0	(p) 93.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	62.8	65.6	69.7	77.0	82.7	85.8	85.5	84.4	82.8	79.0	72.8	65.6		
	DBStd	9.52	6.56	7.39	5.96	5.20	4.31	3.34	3.12	2.92	3.04	4.02	5.27	5.66		
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	294	124	80	23	1	0	0	0	0	0	0	0	0		
	CDD50	9557	398	437	612	809	1012	1074	1100	1067	984	898	683	483		
	CDD65	4374	57	96	169	360	547	624	635	602	534	433	238	79		
	CDH74	44856	126	321	637	2692	6095	8125	8134	7263	5960	3866	1375	262		
(13)	CDH80	18386	9	46	101	715	2564	4062	3889	3132	2320	1220	298	30		
(14)	Wind	WSAvg	5.6	5.7	6.2	6.0	6.4	5.8	5.8	5.1	4.9	5.0	5.2	5.3		
(15) Precipitation	PrecAvg	65.0	0.8	1.1	1.7	3.9	7.3	9.8	10.4	11.9	9.5	5.9	2.3	0.6		
	PrecMax	88.4	3.0	3.6	10.2	10.6	21.7	20.6	19.4	26.2	22.1	18.5	10.2	2.1		
	PrecMin	40.7	0.0	0.2	0.2	0.4	1.1	1.6	4.0	1.5	1.7	0.4	0.0	0.0		
	PrecStd	11.4	0.7	0.9	1.7	2.2	4.0	4.3	4.1	5.3	4.2	4.8	2.3	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.9	84.5	86.4	93.6	99.8	100.6	99.0	97.0	95.1	91.8	89.1	83.8		
		MCWB	71.7	74.0	76.2	81.1	81.4	81.0	81.8	81.4	79.6	76.3	75.9	72.6		
	2%	DB	77.3	80.7	82.6	89.3	95.3	97.6	96.5	94.8	93.1	89.9	86.0	80.2		
		MCWB	69.6	72.2	74.8	79.1	81.6	81.2	81.7	81.4	79.9	76.1	74.2	70.9		
	5%	DB	74.8	77.6	79.6	86.4	92.9	95.2	94.6	93.0	91.3	88.1	83.8	76.9		
		MCWB	68.1	71.0	73.6	78.3	81.3	81.2	81.5	81.4	79.4	75.6	72.9	69.2		
	10%	DB	71.8	75.2	77.2	84.2	90.0	93.2	92.6	91.2	89.6	86.3	80.8	74.0		
		MCWB	66.8	70.5	72.8	77.1	80.1	81.5	81.3	81.3	78.4	75.4	71.5	67.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	75.5	78.3	82.7	85.3	85.9	86.0	86.0	84.6	81.1	78.8	75.4		
		MCDB	77.8	80.3	82.0	88.0	91.6	92.3	92.0	90.5	89.2	86.0	84.0	79.2		
	2%	WB	71.4	73.8	76.3	80.8	83.6	84.5	84.5	84.4	82.8	79.6	77.2	73.4		
		MCDB	75.5	78.2	80.2	86.0	90.3	90.7	90.6	89.5	87.6	84.6	82.2	77.3		
	5%	WB	69.8	72.6	74.9	79.3	82.6	83.6	83.5	83.2	81.8	78.7	76.0	71.4		
		MCDB	73.1	76.6	78.4	84.3	89.1	89.8	89.5	88.6	86.7	83.6	80.6	75.4		
	10%	WB	68.2	71.2	73.7	78.3	81.5	82.7	82.6	82.3	80.8	77.8	74.6	69.3		
		MCDB	71.3	74.8	76.8	82.9	87.7	88.9	88.5	87.9	85.8	82.8	79.0	73.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.4	9.0	10.7	13.1	13.6	12.6	12.1	12.3	12.7	12.8	11.8		
		MCDBR	13.6	12.4	11.8	14.0	16.6	16.5	15.6	14.9	14.8	14.9	15.4	14.3		
		MCWBR	6.8	5.8	6.1	6.8	7.0	6.3	6.4	6.2	5.8	5.5	6.4	6.7		
		MCDBR	11.6	11.1	10.5	12.3	14.7	14.3	13.6	13.1	12.7	13.2	13.3	11.9		
	5% WB	MCWBR	7.2	5.9	6.4	7.1	7.3	6.7	6.8	6.9	6.4	6.0	6.9	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.610	0.642	0.816	0.776	0.566	0.521	0.493	0.553	0.607	0.712	0.614	0.641		
	taud		1.718	1.695	1.426	1.529	2.015	2.157	2.230	2.037	1.862	1.613	1.760	1.680		
	Ebn at Noon		210	214	185	196	240	249	256	243	227	194	207	195		
	Edh at Noon		69	75	102	93	56	49	45	55	65	80	66	70		
(44) All-Sky Solar Radiation	RadAvg		718	855	944	1213	1521	1587	1538	1465	1390	1197	1042	810		
	RadStd		156	142	108	127	146	129	133	111	97	91	135	127		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.16	+1.39	N/A	N/A	N/A	N/A	N/A	+198
(47) Regional (6 neighbors)	N/A	N/A	N/A	+1.04	+0.32	N/A	N/A	N/A	N/A	+144

Nomenclature: See separate page